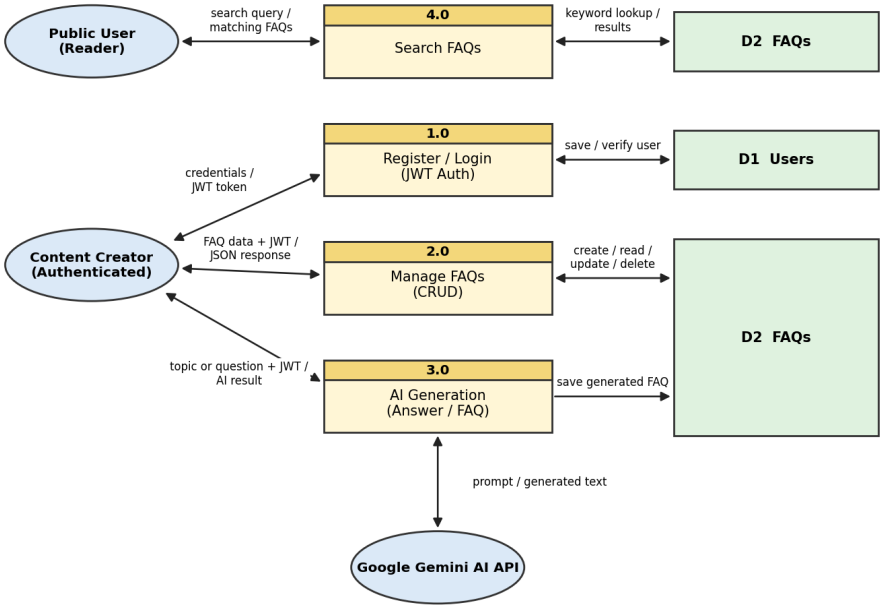
Data Flow Diagram & User Stories

Date	16 September 2026
Team ID	SWTID-2026-3976
Project Name	AI FAQ Assistant API (AnswerBot)
Maximum Marks	4 Marks

Data Flow Diagrams:

See the Data Flow Diagram in this folder (Level-1 DFD).

DFD Level-1: AI FAQ Assistant API



User Stories

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Content Creator / User	Project Setup	USN-1	As a developer, I can set up the Node.js/Express project with dotenv, cors, MongoDB connection and the src folder structure.	Works as described and is verified	High	Sprint-1
Content Creator / User	Registration	USN-2	As a user, I can register by entering my name, email and password, which is validated and stored with a bcrypt-hashed password.	Account created and JWT returned	High	Sprint-1
Content Creator / User	Login	USN-3	As a user, I can log in with email and password and receive a JWT token.	Valid credentials return a token; invalid return 401	High	Sprint-1

User Type	Functional Requirement (Epic)	User Story Number	User Story / Task	Acceptance criteria	Priority	Release
Content Creator / User	Profile	USN-4	As a logged-in user, I can view my profile through a JWT-protected route.	Profile returned only with a valid Bearer token	High	Sprint-1
Content Creator / User	Error Handling	USN-5	As a client, I receive clean, consistent error responses (duplicate email, validation, invalid ID) from a central error handler.	Works as described and is verified	High	Sprint-1
Content Creator / User	Validation	USN-6	As a client, my requests are checked by validation middleware before reaching the controllers.	Works as described and is verified	Medium	Sprint-1
Content Creator / User	Data Models	USN-7	As a developer, I have User and FAQ Mongoose schemas with required fields, category enum and createdBy reference.	Works as described and is verified	High	Sprint-1
Creator / Reader	FAQ Management	USN-8	As a creator, I can create an FAQ with question, answer and category.	FAQ saved with the author reference	High	Sprint-2
Creator / Reader	FAQ Management	USN-9	As any user, I can list all FAQs and open one FAQ by its ID.	Works as described and is verified	High	Sprint-2
Creator / Reader	FAQ Management	USN-10	As a creator, I can update or delete only the FAQs I authored.	Only the author can update or delete	High	Sprint-2
Creator / Reader	Search	USN-11	As a reader, I can search FAQs by keyword using /api/faqs/search?q=.	Matching FAQs returned as JSON	High	Sprint-2
Creator / Reader	AI Integration	USN-12	As a developer, I have a Gemini service that wraps the @google/genai SDK.	Works as described and is verified	High	Sprint-2
Creator / Reader	AI Integration	USN-13	As a logged-in user, I can ask a question and get an AI-generated answer.	AI answer returned with status 200	High	Sprint-2
Creator / Reader	AI Integration	USN-14	As a creator, I can give a topic and receive an AI-generated FAQ (question, answer, category).	Generated question, answer and category returned	High	Sprint-2
Team	Testing	USN-15	As a team, we test all endpoints in Postman and keep a Postman collection.	Works as described and is verified	High	Sprint-3
Team	Testing	USN-16	As a team, we add a test_api.js script and fix issues found during testing.	Works as described and is verified	Medium	Sprint-3
Team	Documentation	USN-17	As a team, we document architecture, ER diagram, user flow and the API testing specification.	Works as described and is verified	High	Sprint-3
Team	Demo	USN-18	As a team, we prepare and rehearse the project demonstration.	Works as described and is verified	Medium	Sprint-3
Team	Performance	USN-19	As a team, we record response times of key endpoints.	Works as described and is verified	Medium	Sprint-3